

wisp



Wireless Intelligent Sensing for Personalized Self-Triage

Progress report v2 — the patient-first redesign, the real ESP32 sensor, and what is still open.

When “I just feel weak” is all you can say, WISP can check whether your physical function is actually different from your usual self, then tell you what to do next and why.

Status as of 6 October 2026 · branch **refactor/product-v2** on github.com/FerosC101/WISP

Screens captured from the running system. Patient-flow screenshots use recorded (synthetic) sensor replay, labelled in the app; the live-sensor screenshot shows real data from the team’s ESP32-S3.

1. Summary

WISP is a calm self-triage assistant. A person says how they feel; WISP asks a few short safety questions, decides whether a quick physical check would actually change the advice, and ends with a clear next step: **what to do, by when, and why**. Wi-Fi sensing is one optional tool the agent uses only when it matters — never the product itself.

Version 2 rebuilds the experience around that idea: **simple on top, deep underneath**. The patient sees a conversation and a recommendation; judges and engineers get separate views of the agent's decisions and the sensor.

112 / 112

automated backend tests
passing

3 / 3

demo scenarios run end-to-end
in the browser, desktop
and 390 px phone

15 / 15

test conversations: right care
level and right measure /
don't-measure choice

~100 / s

real CSI readings streamed
from the ESP32-S3 into the
app

What changed in v2

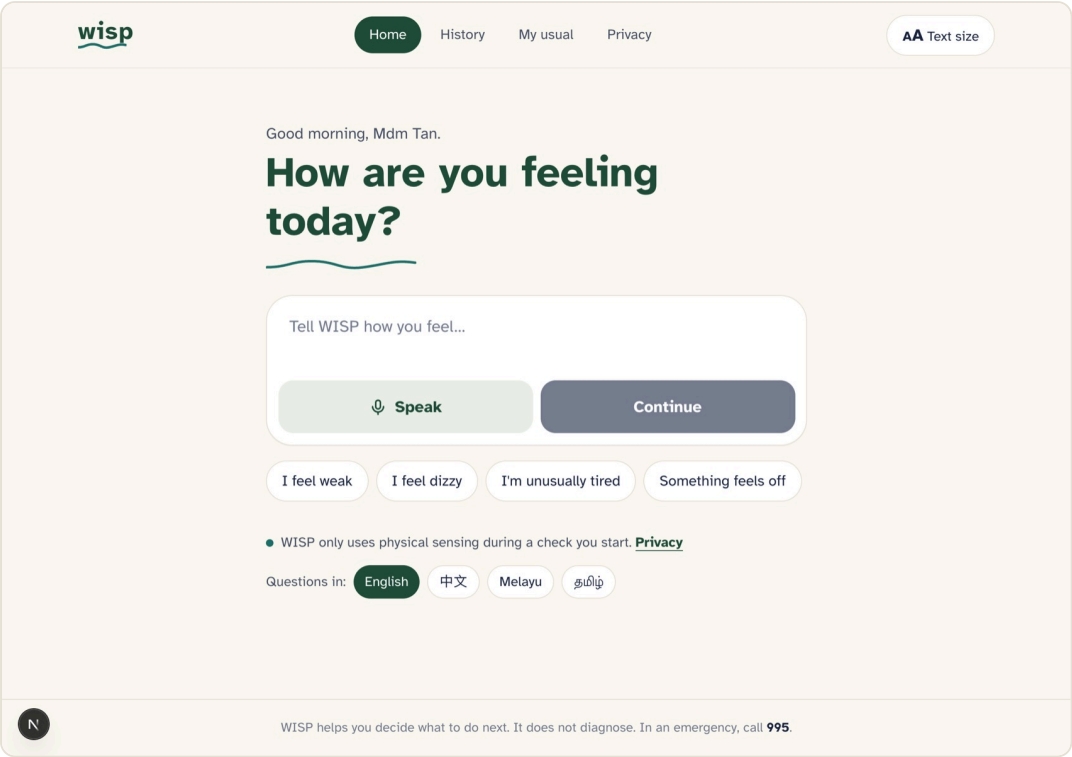
- **Three separate views:** the patient app (default), a **Technical view** with the structured decision record for judges, and an **Engineering view** for sensing and demo control. Internal details no longer leak into the patient screens.
- **New identity:** forest green, warm ivory and sage, with “the WISP line” motif; red only for emergencies; mobile-first with a bottom tab bar.
- **New home:** “How are you feeling today?” with text or voice and quick prompts; demo profiles moved to demo mode.
- **Visible agentic choice:** “A short movement check could help — Do the check / Skip”, with a plain-language “Why this check?”.
- **Recommendation as the hero**, with a human-readable “How WISP decided”.
- **Safety questions in English, Mandarin, Malay and Tamil**, answered with buttons whose values are language-independent.
- **Real hardware:** WISP's own ESP32-S3 firmware (built in CI), Wi-Fi set up from the app, and a live sensor monitor.
- **Clinician review pack**, real-trial collection tooling and a WorkBuddy live-connection checklist.

The deterministic safety architecture is unchanged: red-flag screen before sensing; any warning sign → emergency and sensing locked; sensing can raise urgency but never lower it below the symptom floor; unreliable readings are never used; the AI cannot fabricate measurements.

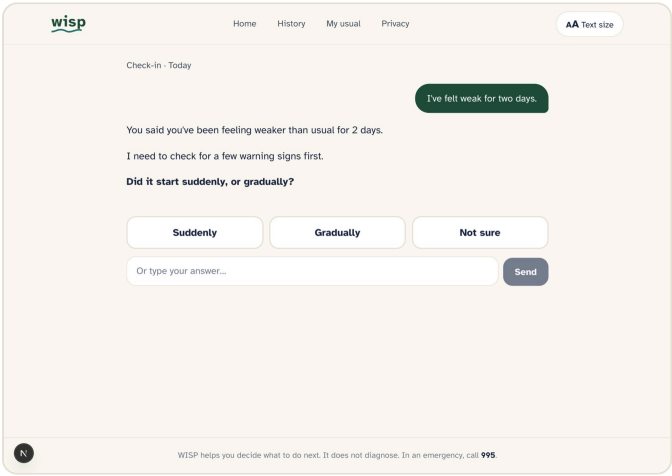
2. Status

AREA	WHERE IT STANDS	STATUS
Patient app v2	Home, conversation, movement-check offer, quick movement check, recommendation, How WISP decided, history timeline, My usual, privacy. Works at 390 px.	Done
Technical & Engineering views	/explain/<session> structured record + tool-call timeline (live); /dev with live sensor, demo launchers, audit log, WorkBuddy filter.	Done
Safety rules engine	Unchanged and fully tested; patient wording updated to “movement check”.	Done
Multilingual safety questions	EN / / Melayu / தமிழ் built and tested. Translations are drafts.	Needs native review
ESP32 firmware	WISP firmware 1.2 built in GitHub Actions, flashed to the ESP32-S3 (original firmware backed up). Joins Wi-Fi set from the app; auto-corrects network names.	Done
Real sensing	Real CSI streaming at ~100 readings/s, 52 usable subcarriers, visible live in the app.	Streaming
Real 5-times sit-to-stand	Not yet performed in front of the board. Pipeline thresholds come from synthetic data and may need re-tuning.	Needs a test
Real accuracy numbers	Collection script and report ready; 0 real trials so far.	Needs participants
Tencent WorkBuddy	MCP server verified with a scripted client; 8-point live checklist written. Not yet run against real WorkBuddy.	Needs WorkBuddy
Clinical review	docs/clinical_review.md ready with review log. Not yet reviewed — logic must be described as “prototype”.	Needs a clinician

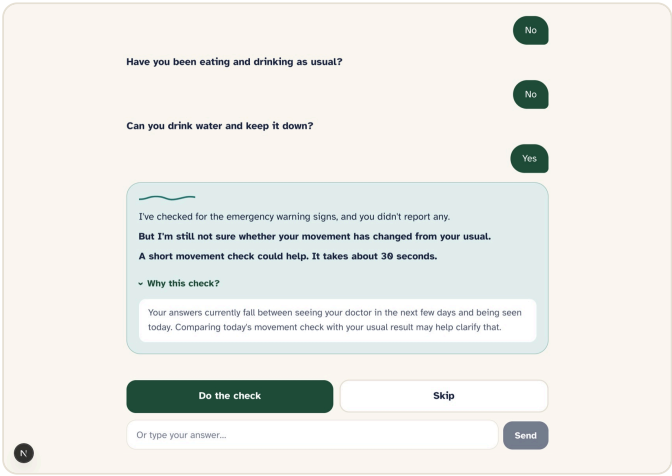
3. The patient experience



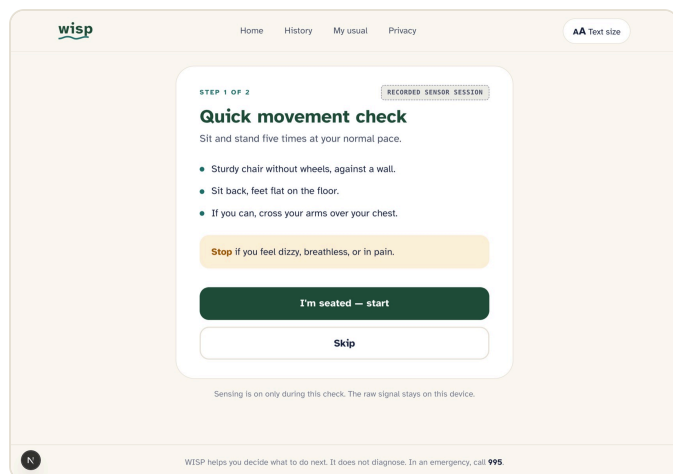
Home. A greeting and one question — “How are you feeling today?” — with text or voice input, quick prompts and language choice for the safety questions. No demo profiles, no sensing language.



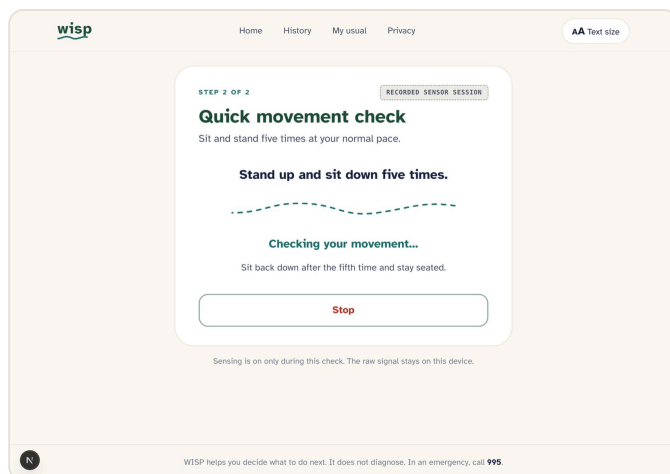
Conversation. WISP repeats the complaint back, says it needs to check warning signs, then asks one short question at a time with large answer buttons.



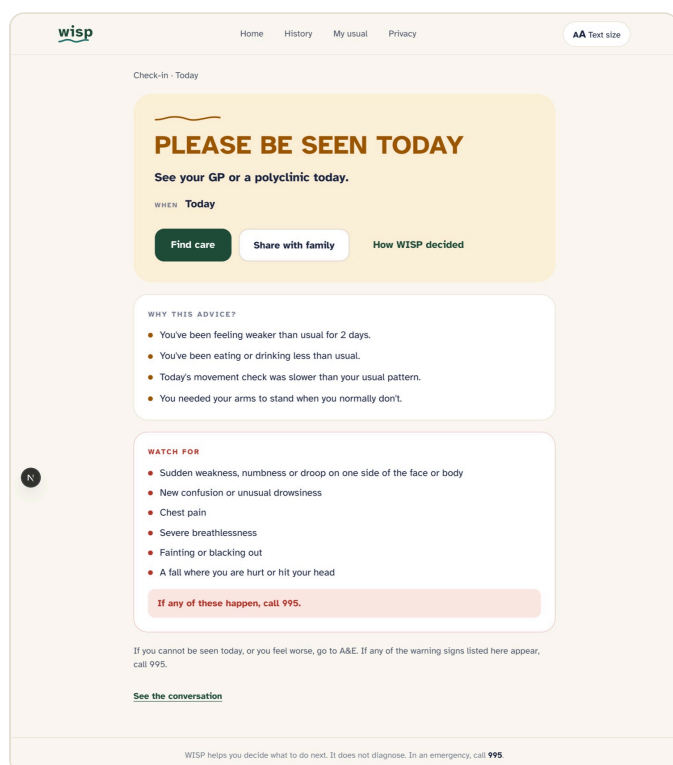
The agent's choice, made visible. “A short movement check could help.” *Why this check?* explains the uncertainty in plain words. Do the check / Skip.



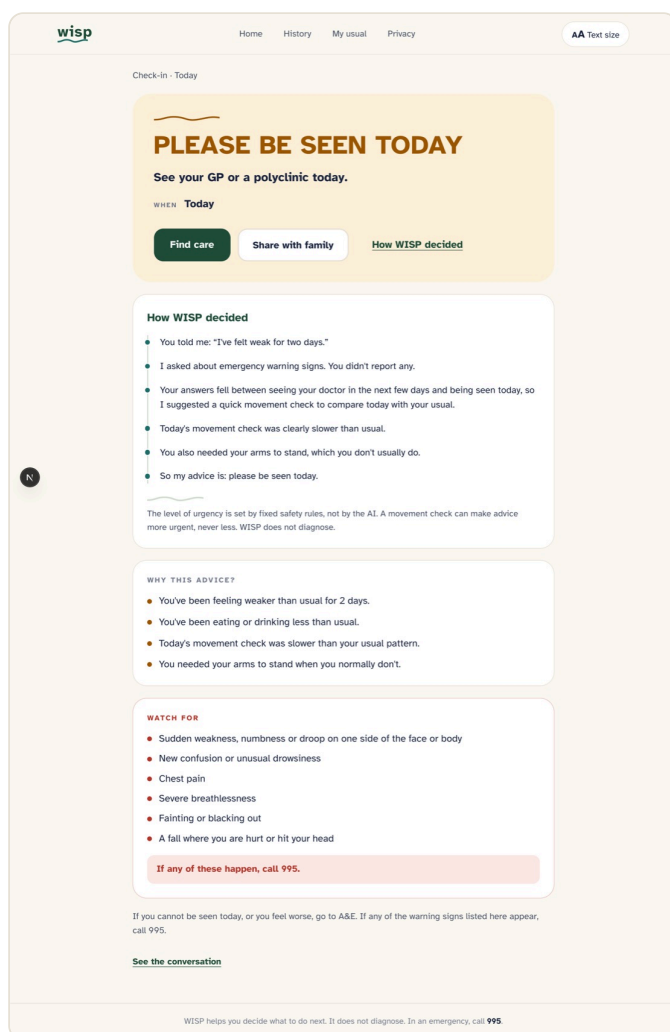
Quick movement check. “Sit and stand five times at your normal pace.” Simple setup and stop advice; replay is clearly labelled.



Checking your movement... The WISP line moves; no waveforms, timings or confidence scores are shown to the patient.

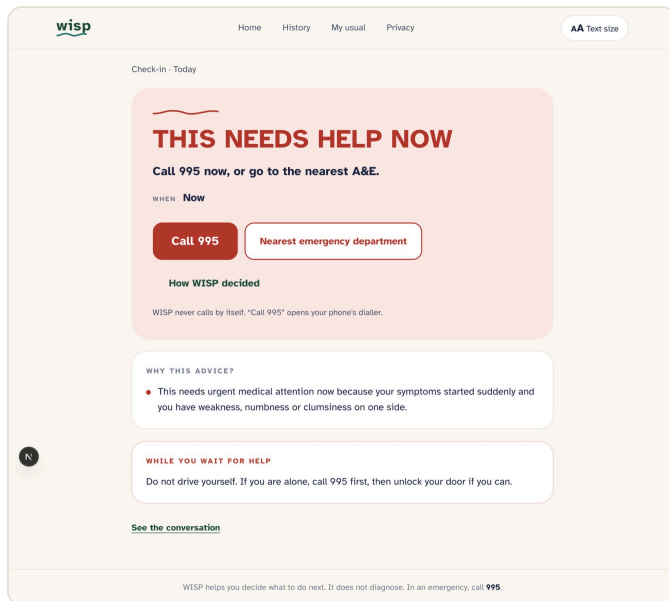


Scenario 1 — Please be seen today. The action dominates the screen: what, when, why, Find care, Share with family, and warning signs.

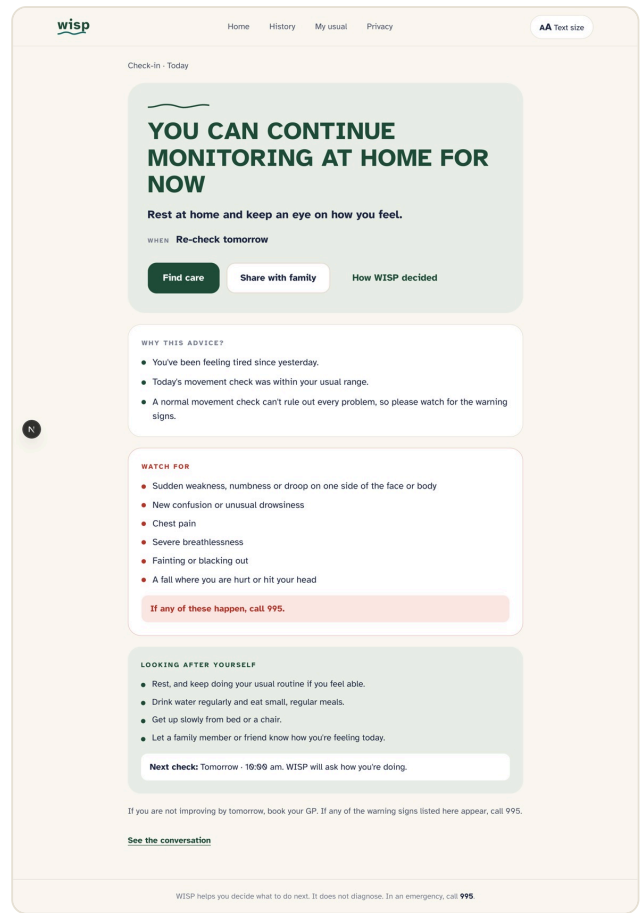


How WISP decided. A plain-language account built from the structured record: what you said, what was checked, why the movement check was offered, what it showed.

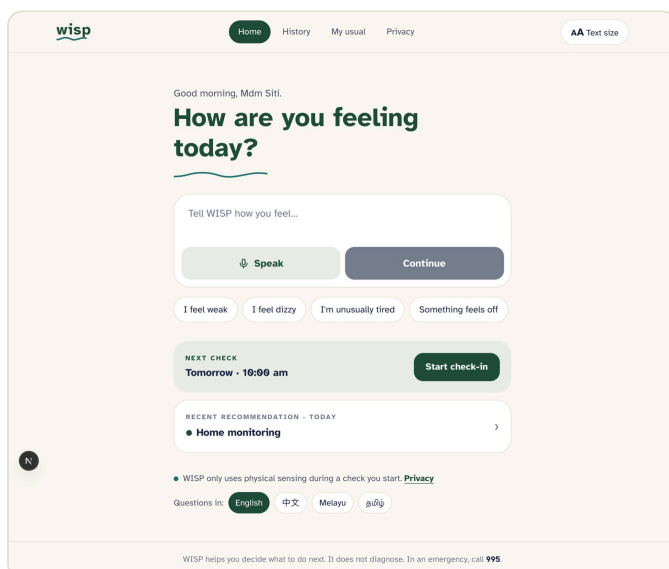
4. Demo scenarios



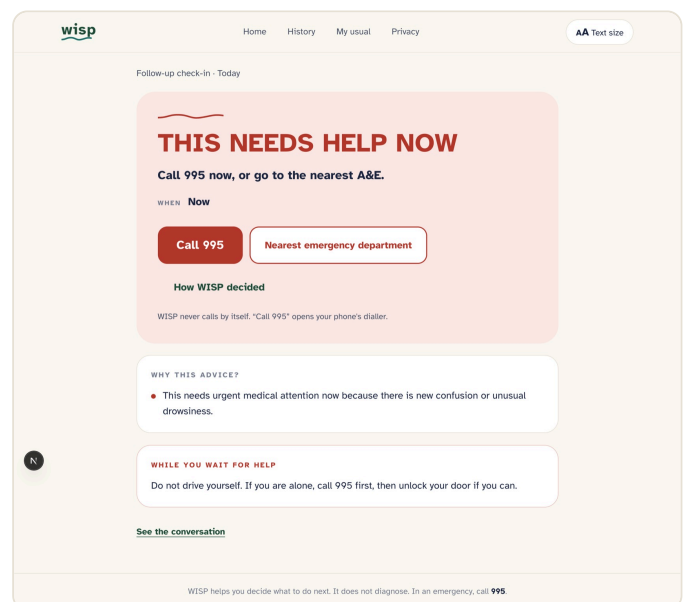
Scenario 2 — the agent refuses to sense. “I suddenly felt dizzy and my left hand feels clumsy” → This needs help now. Call 995. No movement check is offered.



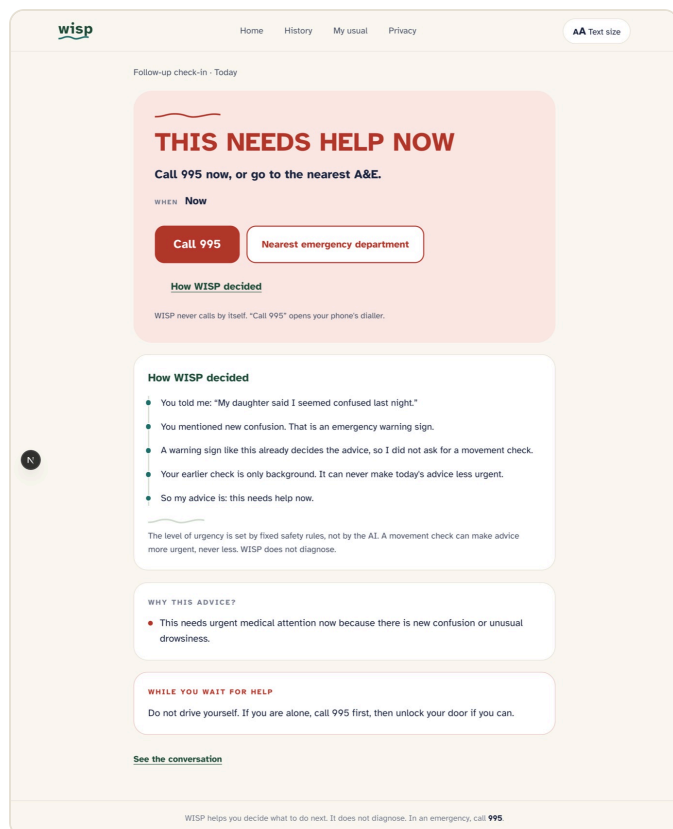
Scenario 3, day 1 — home monitoring. Movement within her usual range; self-care advice and a check-in booked for tomorrow 10 AM.



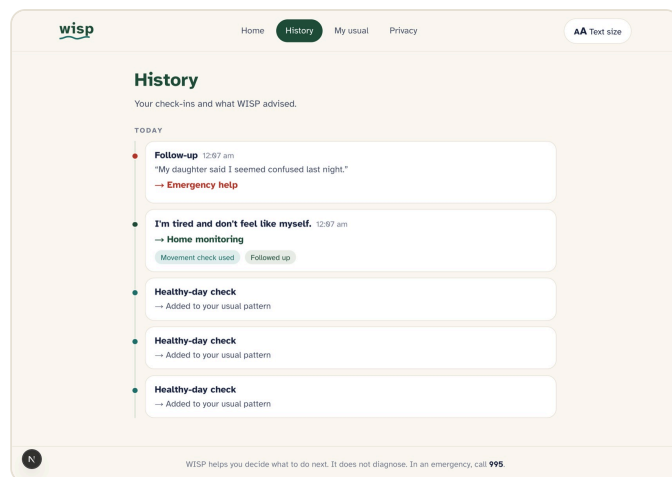
Next check on the home screen. The scheduled check-in and the most recent recommendation.



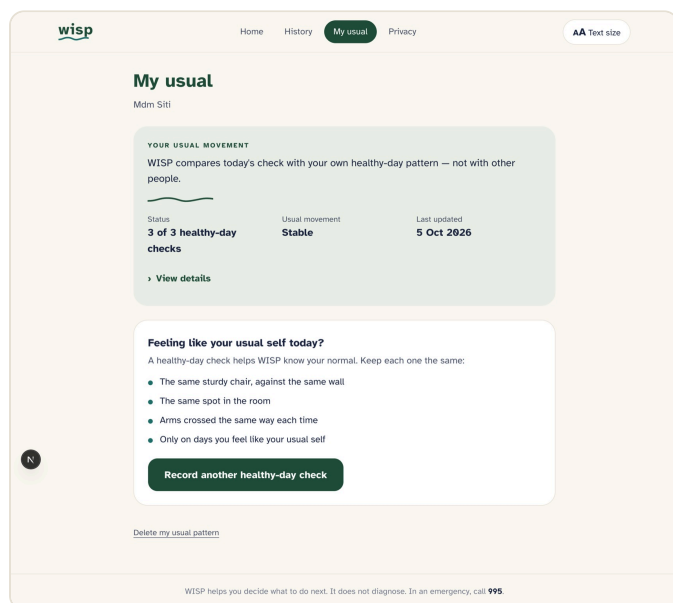
Scenario 3, day 2. “My daughter said I seemed confused last night” → emergency. Yesterday's normal result never lowers today's urgency.



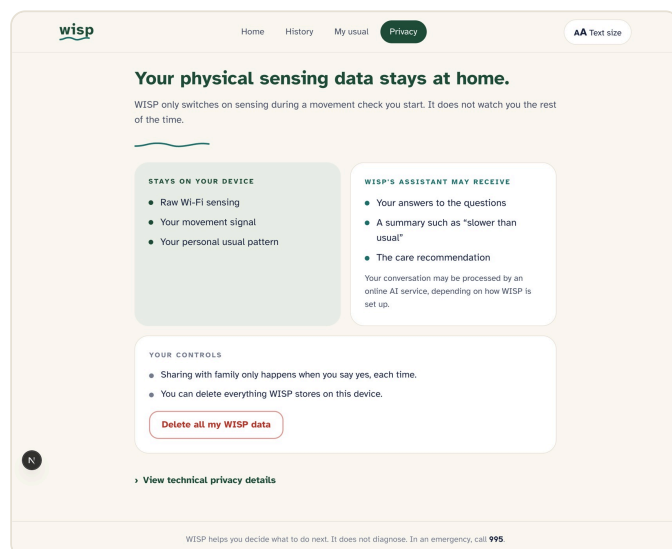
How WISP decided (day 2). “Your earlier check is only background. It can never make today’s advice less urgent.”



History. A day-by-day timeline: complaint, outcome, whether a movement check was used, follow-up status, healthy-day checks.



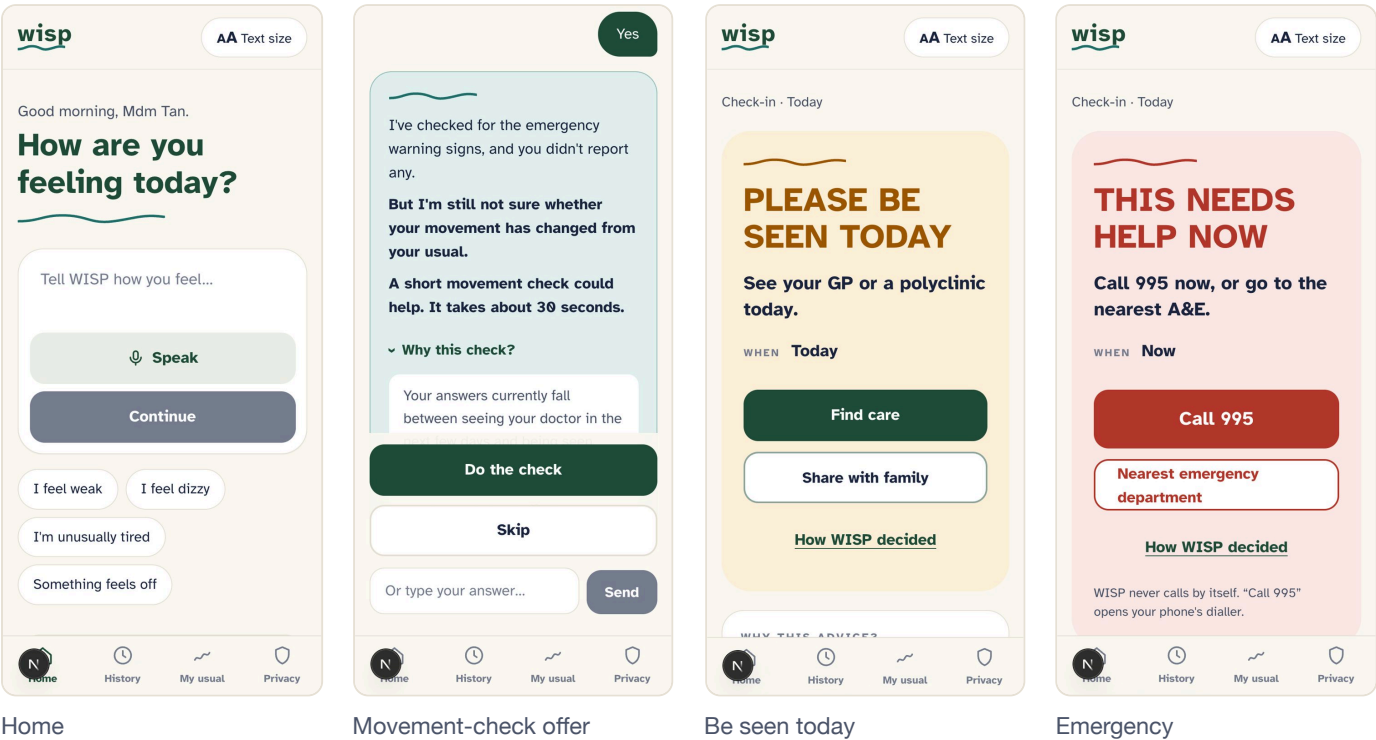
My usual. Healthy-day checks recorded, “Stable”, last updated; seconds only behind “View details”.



Privacy. “Your physical sensing data stays at home.” What stays on the device vs what WISP’s assistant may receive; technical diagram on request.

On a phone (390 px)

The patient app is designed mobile-first: full-width buttons, large touch targets, a bottom tab bar, and the urgent action visible without scrolling.



5. For judges: the Technical view

Opened from the demo-mode bar (*Technical view*) or at /explain/<session>. It is a structured decision record, not model chain-of-thought, and it updates live — useful on a second screen during the demo.

The screenshot shows the 'wisp' interface for a 'check-in' session with 'Mdm Tan'. The top navigation bar includes 'Home', 'History', 'My usual', 'Privacy', and a 'AA Text size' button. The main header indicates 'TECHNICAL VIEW · STRUCTURED DECISION RECORD' and 'Physical sensing COMPLETE'. The patient's name 'Mdm Tan · check-in' is displayed, along with the agent information: 'Agent: built-in agent (offline stand-in) · session s_8db843b083536 · not model chain-of-thought'.

The decision record is organized into several sections:

- 1 CURRENT CONCERN:** "I've felt weak for two days."
- 2 SAFETY SCREEN:** "Passed — no emergency warning signs" (deterministic rule).
- 3 POSSIBLE CARE RANGE:** "Primary care soon → Same-day care" (floor T3, ceiling T2).
- 4 MISSING INFORMATION:** "None outstanding".
- 5 OPTIONS CONSIDERED:** "Ask another question", "Physical function check", "Recommend care now".
- 6 SELECTED ACTION - WHY:** "Physical function check" (AI agent / deterministic rule). "A functional measurement could help distinguish primary care soon from same-day care."
- 7 TOOL CALLED:** "run_functional_assessment("5xSTS")".
- 8 SENSOR RESULT:** "5 rises · total 15.8 s · per rise 2.47 / 2.9 / 3.16 / 3.64 / 3.61 s". "measurement conf 1.00 · single-person 1.00 · wisp-synthetic-csl · verified true · arms used (self-report)". "RECORDED SENSOR SESSION · SYNTHETIC".
- 9 BASELINE RESULT:** "Clearly slower than usual · slower_than_usual · severity clear · confidence high". "The total time was outside this user's usual recorded range and the user required arm support not normally used."
- 10 CARE-TIER CHANGE:** "Primary care soon → Same-day care".
- 11 FINAL DISPOSITION:** "T2 · Same-day care" (deterministic rule, confidence high). "SR-3 Concerning non-emergency finding(s): reduced_intake → Symptom floor T3, ceiling T2". "FN-3b 5xSTS clearly slower than usual + concerning finding → Functional tier T2". "ESC-1 Functional evidence raised urgency T3 → T2 → Tier T2".

A right-hand sidebar titled 'TOOL CALLS & DECISIONS' lists the sequence of tool calls and decisions with timestamps and agent information.

At the bottom, a footer states: "WISP helps you decide what to do next. It does not diagnose. In an emergency, call 995".

Scenario 1. Safety screen passed → range Primary care soon → Same-day care → selected *Physical function check* with reason → run_functional_assessment("5xSTS") → sensor and baseline result → **care-tier change Primary care soon → Same-day care** → rules SR-3, FN-3b, ESC-1. Right: tool calls by actor (AI agent / deterministic rule / sensor / patient).

The screenshot shows the 'wisp' interface for a 'check-in' session with 'Mr Lim'. The top navigation bar is identical to the previous screenshot. The main header indicates 'TECHNICAL VIEW · STRUCTURED DECISION RECORD' and 'Physical sensing LOCKED'. The patient's name 'Mr Lim · check-in' is displayed, along with the agent information: 'Agent: built-in agent (offline stand-in) · session s_7affb881ecdff · not model chain-of-thought'.

The decision record is organized into several sections:

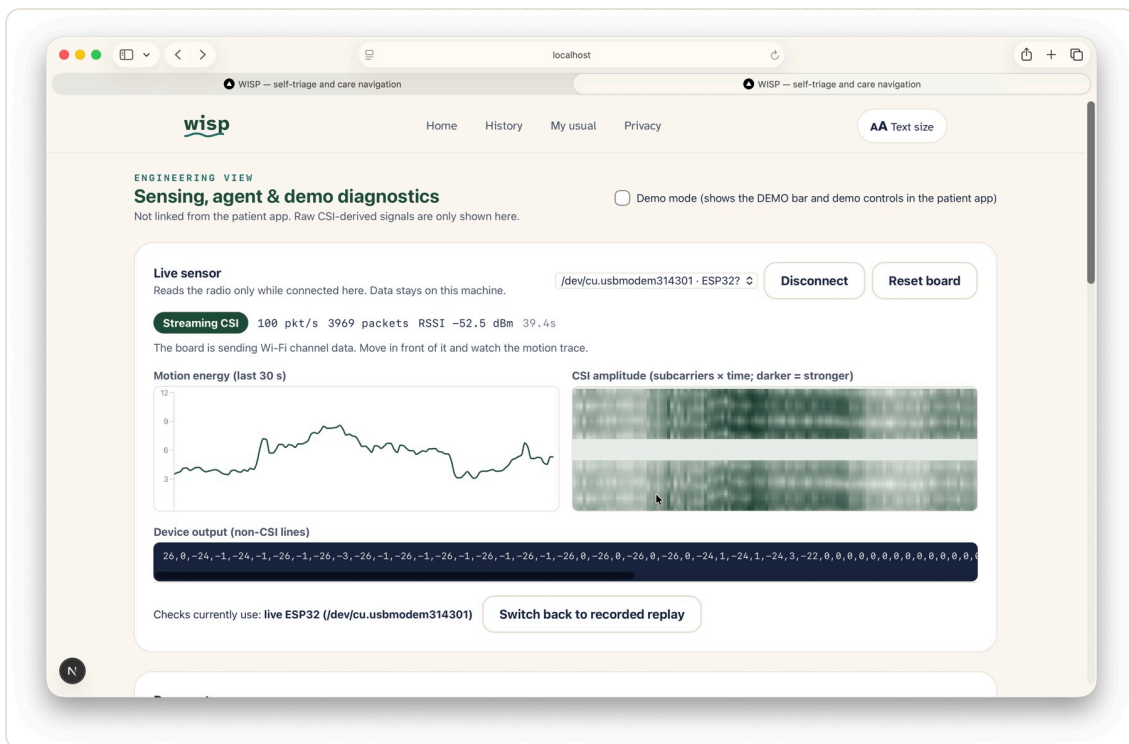
- 1 CURRENT CONCERN:** "This morning I suddenly felt dizzy and my left hand feels clumsy."
- 2 SAFETY SCREEN:** "Triggered: Sudden onset, One-sided weakness" (deterministic rule).
- 3 POSSIBLE CARE RANGE:** "Emergency" (floor T1, ceiling T1).
- 4 MISSING INFORMATION:** "None outstanding".
- 5 OPTIONS CONSIDERED:** "Ask another question", "Physical function check (locked)", "Recommend care now".
- 6 SELECTED ACTION - WHY:** "Escalate now" (AI agent / deterministic rule). "Emergency warning sign reported. Physical sensing not requested: It cannot change an emergency recommendation."
- 7 TOOL CALLED:** "No physical assessment requested".
- 8 SENSOR RESULT:** "—".
- 9 BASELINE RESULT:** "—".
- 10 CARE-TIER CHANGE:** "Sensing not requested: emergency warning sign already determines the recommendation".
- 11 FINAL DISPOSITION:** "T1 · Emergency" (deterministic rule, confidence high). "RF-1 Emergency warning sign reported: Sudden onset, One-sided weakness / numbness → Tier T1: Physical sensing locked for this session."

A right-hand sidebar titled 'TOOL CALLS & DECISIONS' lists the sequence of tool calls and decisions with timestamps and agent information.

At the bottom, a footer states: "WISP helps you decide what to do next. It does not diagnose. In an emergency, call 995".

Scenario 2. Safety screen triggered (sudden onset, one-sided weakness); *Physical function check* struck through; **SENSING NOT REQUESTED — emergency warning sign already determines the disposition**; sensing locked; no sensor tool call.

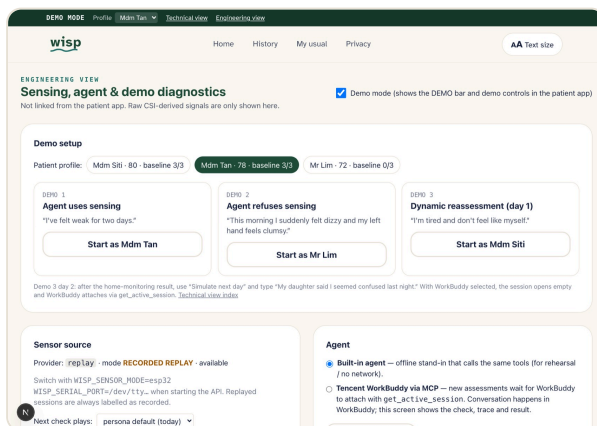
6. Real sensing: the ESP32-S3 and the Engineering view



Live sensor — real data. The team's ESP32-S3 joined the Wi-Fi, pings the router ~100 times per second and streams channel state information: 100 readings/s, RSSI -52.5 dBm. The motion trace rises with movement near the board; the heatmap shows 64 subcarriers over time (the pale band is the unused centre subcarriers).

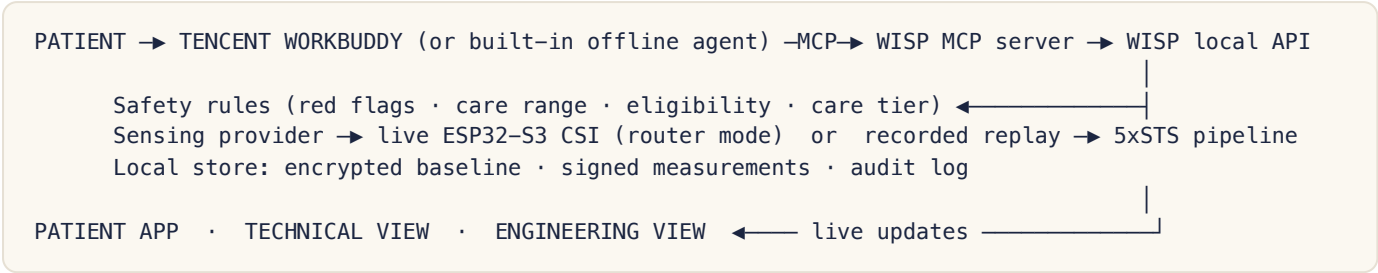
How the board got here

- It was running another project's firmware ("HARVESTGUARD POD"). A full 16 MB backup was taken before anything was changed (restorable with one command).
- **WISP firmware** (based on Espressif's `csi_recv_router`) is built by GitHub Actions — no local toolchain needed — and flashed with `esptool`.
- Wi-Fi details are entered in the app and sent over USB to the board only; they are never in code, builds or GitHub.
- When it couldn't join, the board's own scan showed the router's real name ended with a space; firmware 1.2 now corrects names like this automatically.



Engineering view (/dev). Demo mode switch, one-click demo scenarios, patient profile, sensor source, plus (further down) the live sensor, measurement plots, ground-truth entry and the audit log with a WorkBuddy filter.

7. How it is built



AI AGENT (WORKBUDDY)	FIXED SAFETY RULES (CODE)
Holds the conversation, structures the case, notices missing information, chooses whether a movement check is worth doing, explains the result.	Emergency screening, permission to sense, care floor and ceiling, final care level, abstention. The AI cannot skip, reorder or override them.

8. Evaluation so far

All figures below are on synthetic data with small samples. No real-world accuracy or population-level fairness is claimed.

EVALUATION	RESULT
End-to-end test conversations (v2 flow)	15/15 correct care level and measure / don't-measure choice
Fairness: only age, sex, living situation, language preference or phrasing changed	0 of 51 variants changed the result
Warning signs found in free text (each is also asked directly)	14/15, 0 false alarms; missed a Malay-only sentence
Movement-check timing on synthetic Wi-Fi data (80 sessions)	Average error 0.21 s; 7 wrongly rejected
Second person walking through (60 synthetic sessions)	30/60 rejected — experimental; the main safeguard is asking first
Real participants with stopwatch timing	0 so far — tooling ready

9. What is still needed

Needs the team

1. **Run a real movement check on the ESP32.** Engineering view → *Use this sensor for checks* → Demo 1 → five sit-to-stands with the chair between the board and the router. Expect re-tuning of the five-rise detection (raw data is saved locally for this).
2. **Collect real accuracy data:** 3–5 people × 5–10 tries with a stopwatch, using `scripts/collect_trials.py`; report with `real_report.py`. Only these numbers should be quoted.
3. **Connect Tencent WorkBuddy** and complete the 8-point checklist in `docs/workbuddy/README.md` for scenarios 1 and 2.

Before any real patient use

- **Clinical review** of the rules using `docs/clinical_review.md`.
- **Native-speaker review** of the Mandarin, Malay and Tamil safety questions.

Housekeeping

- Merge `refactor/product-v2` into `main` (`main` still holds `v1`).
- Keep the live-ESP32 setting across API restarts (it currently resets to recorded replay).
- Optional: translate the recommendation screens; improve second-person detection (e.g. a second receiver).

Run locally with `./scripts/demo.sh` (add `--mcp` for WorkBuddy) and open `http://localhost:3000`. The demo script is in `docs/demo.md`.